MILITARY
SPECIFICATION

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MILITARY SPECIFICATION

RECORDER, RADIOSONDE AN/TMQ-5( )
(Units of)

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1. SCOPE

1.1 This specification covers the following units of one type of equipment designated Recorder, Radiosonde AN/TMQ-5( ).

Cabinet, Electrical Equipment CY-1390( )/TMQ-5
Cabinet Subassembly, Electrical Equipment MX-1482( )/TMQ-5
Case, Accessories CY-930( )/TMQ-5
Converter, Signal Data CV-146( )/TMQ-5
Panel, Control C-834( )/TMQ-5
Paper, Graph ML-432( )/TMQ-5
Power Supply PP-968( )/TMQ-5
Recorder, Frequency-Time RD-88( )/TMQ-5

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton Wrapping
QQ-S-781	Steal Strapping, Flat
PPP-T-45	Tape: Paper, Gummed, Reinforced
PPP-B-601	Box, Wood, Cleated-plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, corrugated and solid, sheet stock
PPP-P-291	Paperboard, wrapping, cushioning
PPP-T-76	Tape, pressure-sensitive adhesive paper, water resistant

FSC 6660

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MIL-E-75	Electron tubes, packaging, packing and container marking of.
MIL-P-116	Preservation, methods of
MIL-P-11268	Parts, materials and processes used in communication equipment
MIL-M-13231	Marking of electronic items
MIL-F-14072	Finishes for ground signal equipment
MIL-C-45662	Calibration of standards
MIL-H-15424	Hand Tools, Packaging of

STANDARDS

MIL-STD-105	Sampling procedures and tables of inspection by attributes
MIL-STD-129	Marking for shipment and storage
MIL-STD-169	Extreme temperature cycle
MIL-STD-170	Moisture resistance test cycle
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Military Standard Environmental Test Methods for Aerospace and Ground Equipment

DRAWINGS

SC-DL-125001	Radiosonde Recorder AN/TMQ-5( )
SC-DL-125700	Cabinet, Elect. Equip. CY-1390( )/TMQ-5
SC-DL-125400	Cabinet, Sub-Assy. Elect. Equip. MX-1482( )/TMQ-5
SC-DL-125800	Case, Accessories CY-930( )/TMQ-5
SC-DL-125300	Converter, Signal Data CV-146( )/TMQ-5
SC-DL-125100	Panel, Control C-834( )/TMQ-5
SC-DL-125200	Power Supply PP-968( )/TMQ-5
SC-DL-125500	Recorder, Freq. Time RD-88( )/TMQ-5
SM-D-347544	Paper, Graph ML-432( )/TMQ-5

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The Radiosonde Recorder AN/TMQ-5( ) is used to record audio modulated signals of meteorological data transmitted from balloon borne radiosondes.

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3.2 Preproduction samples. - The contractor shall furnish three preproduction samples for approval, if required by the invitation for bids and contract. (See 6.1(f)(1))

3.3 Construction. - The equipment shall be constructed in accordance with the requirements of this specification, and the drawings of para. 2.1. The requirements of MIL-P-11268 shall apply when the part, material or process is not specified by drawing.

3.4 Finish. - The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4)

3.5 Marking. -

3.5.1 General. - Marking shall conform to the equipment drawings and MIL-M-13231. (See 4.4)

3.5.2 Serial Numbers. - The items listed in 1.1 shall have serial numbers except Graph Paper ML-432.

3.6 Interchangeability. - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.7 Performance. -

3.7.1 Recorder, Frequency-Time RD-88( )/TMQ-5. -

3.7.1.1 Signal voltage range. - The voltage measuring range and print out of the recorder shall conform to 3.7.3.4.

3.7.1.2 Speed of response. - The pen shall record 95 divisions, ± 0.15 division, in the time and under the conditions stipulated in 4.9.1.1.

3.7.1.3 Linearity. - The recorder shall be adjusted so the record will have a linear-relation to the proportional d.c. voltage applied at the input of the recorder under the standard test conditions specified in 4.9.1. The deviation from linearity throughout the range of 5 to 95 recorder divisions shall not exceed 0.15 division of the full scale, including the deviations contributed by hysteresis specified in 3.7.1.5.

3.7.1.4 Stability. - With a constant d.c. input voltage corresponding to any recorder position between 5 and 95 scale divisions, and with variations in line voltage between 105 and 125 volts and variations in line frequency between 50 and 65 Hz, the recorder record shall not deviate from the record produced at the start of the test when operating with a line voltage of 115 volts and line frequency of 60 Hz by more than 0.05 percent during a 30-minute period of operation.

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3.7.1.5 Hysteresis. - The recording of the d.c. input voltage to the recorder when approaching from up scale shall not deviate from the recording of the same voltage when approached from down scale by more than 0.1% of full scale (±0.5% sensitivity) at any point in the range of the recorder between 5 and 95 recorder divisions, under the standard test conditions of 4.9.1. Under the normal and low temperature conditions of 3.9.2(a), the deviation shall not exceed 0.2% of full scale (± 0.1% sensitivity).

3.7.1.6 Zero adjustment. - The electrical zero adjustment shall have an adjustment range of not less than ± 0.5 percent. The limit of pen travel shall be determined by the position of the mechanical zero stop. The marking assembly shall be capable of being mechanically repositioned on the drive cable.

3.7.1.7 Chatter. - There shall be no chatter of the pen-holder assembly when moving either up or down scale and the pen-holder clamp screws shall not back-off under vibration.

3.7.1.8 Pen lift. - There shall be no "tails" greater than 2 recorder divisions between 60 and 120 Hz when leaving the 60 Hz position, and there shall be no "tails" when approaching the 120 Hz position, when subjected to 4.9.1.4.

3.7.1.9 Pen pressure. - The pressure of the pen point on the chart paper shall be 2 oz. ± 5 grams.

3.7.1.10 Chart feed. - The chart shall operate at preset speeds of 1/2 inch, 1 inch and 2 inches per minute ±0.1%.

3.7.1.11 Torque measurements. - Under all operating conditions the torque shall be 7 to 12 inch pounds at the center (on the locknut) of the maindrive gear, directly beneath the slide wire assembly. With the equipment "off" the drag of the recorder mechanism measured by a spring scale, or satisfactory equivalent, attached to the recorder indicator shall not exceed 30 ounces.

3.7.2 Converter, Signal Data CV-1146( )/TMQ-5. -

3.7.2.1 Audio Frequency Range. - The converter shall measure frequencies in the range from 10 to 220 Hz and shall produce a proportional d.c. output voltage. (See 3.7.3.4)

3.7.2.2 Reference adjust range. - The input signal indication shall be capable of being adjusted to 95 recorder divisions with an input signal frequency range of 152 thru 228 Hz.

3.7.2.3 Linearity. - The deviation from linearity of the converter shall not exceed 0.05 percent of the full scale at any signal frequency within the range of the unit. The linearity does not include the effects of signal amplitude and wave shape specified in 3.7.3.4.

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3.7.2.4 Stability. - With an input signal of constant frequency, amplitude and wave shape, the change inconverter output shall not exceed 0.05 percent of full scale during a 30-minute test interval with line voltage variations between 50 to 65 Hz, under the standard test conditions of 4.9.1 and normal temperature conditions of 3.9.2(a). Under the low temperature operating conditions of 3.9.2(a) the change shall not exceed 0.1 percent of full scale.

3.7.3 Combined requirements for frequency-time recorder and signal data converter. - In order to correlate the performance tests in 4.9 to the requirements of Section 3, the following combined requirements shall apply:

3.7.3.1 Hysteresis. - The recording of an input signal to the converter between 10 and 220 Hz when approached from a high frequency, shall not deviate from the recording of the same signal when approached from a lower frequency by more than 0.1 percent of full scale (±0.05 percent sensitivity) under the standard test conditions of 4.9.1 and normal temperature conditions of 3.9.2(a). Under the low temperature operating conditions of 3.9.2(a) this deviation shall not exceed 0.2 percent of full scale. (±0.1 percent sensitivity)

3.7.3.2 Linearity. - The equipment shall be adjusted so the record will have a linear relation to the input signal frequency within the range from 10 to 220 Hz. Under the standard test conditions of 4.9.1 and normal temperature conditions, 3.9.2(a), the deviation from linearity, including the hysteresis of 3.7.3.1, but excluding the effects of signal amplitude and wave shape specified in 3.7.3.4 shall not exceed 0.2 percent of full scale. Under the low temperature operating conditions of 3.9.2(a) the change shall not exceed 0.25 percent of full scale.

3.7.3.3 Stability. - With an input signal of constant frequency, amplitude and wave shape, the change in recorded data shall not exceed 0.10 percent of full scale during a 30 minute test interval with line voltage variations between 105 and 125 volts and line frequency variations between 50 and 65 Hz, under the standard test conditions of 4.9.1 and normal temperature conditions of 3.9.2(a). Under the low temperature operating conditions of 3.9.2(a) the change shall not exceed 0.15% of full scale.

3.7.3.4 Audio signal voltage and wave form. - The equipment shall operate with any of the following type signals within the specified frequency range: Sine, square, negative-going saw-tooth and negative pulse. The combined effect of variations in signal amplitude between 10 and 75 volts peak to peak and pulse widths between 1000 and 2500 microseconds shall not exceed 0.1 chart paper division from that recorded with a sine wave of 25 volts peak to peak at all corresponding frequencies between 10 thru 220 Hz.

3.7.4 Power Supply PP-968( )/TMQ-5. - The PP-968( )/TMQ-5 power supply shall meet the following requirement.

3.7.4.1 Outputs. -

(a) 300v. Unregulated supply:

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Input	Frequency	External Load	Output
105v	50 - 60 Hz	2500 ohms, ±5%	450v, ±10%
115v	50 - 60 Hz	2500 ohms, ±5%	500v, ±10%
125v	50 - 60 Hz	2500 ohms, ±5%	550v, ±10%

NOTE: The external load shall be applied from the junction of R418 and C409 to ground.

(b) ±300v, ±3% regulated supply: The +300V regulated output shall not vary more than 2% over a load range of 4000 to 25,000 ohms, ±5%, as the input power source is varied from 105v ac to 125v ac at 50 - 60 Hz. At the same time, the ripple content shall not exceed 0.03v max., (peak to peak) with a 12,000 ohm load.

NOTE: The external load shall be applied to test jack J404. (Cathode of the Positive Series Regulator tube, V402).

(c) -175v, ±3% regulated supply. - The -175 volt output shall not vary more than 1% over a load range of 1060 to 1260 ohms, ±5%, as the input power source is varied from 105 to 125v at 50-60 Hz. At the same time, the ripple content shall not exceed 0.1 volt maximum (peak to peak) with a 1160 ±5% ohm load.

NOTE: The external load shall be applied at test jack J413.

3.7.5 Fan operation. - The fan shall conform to 4.9.3..

3.7.6 Speaker. - The speaker output shall be free of audible distortion over the frequency range of the equipment and shall be capable of being heard, in a reasonably quiet room, at a distance of 25 feet.

3.8 Paper, Graph ML-432( )/TMQ-5. - The ML-432( )/TMQ-5 graph paper shall conform to SM-D-347544.

3.9 Service conditions. - The equipment shall meet the following service conditions:

3.9.1 Operation. - Continuous use for a period of 3 years, under the environmental requirements specified in 3.9.2 through 3.9.4, with a duty cycle of 3 hours on and 3 hours off with no more than normal maintenance and replacement of parts. Where it is required to make a selection of parts, materials, processes, construction methods, etc., the contractor shall conform to 3.3. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.9.2 Temperature (See 4.10). -

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(a) Operating. - Ambient temperature in the range of +150°F to -40°F. (The 150°F temperature includes effect on sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.9.3 Relative humidity (See 4.11). - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.9.4 Rain (AN/TMQ-5( ) and CY-930) (See 4.12). - Two inches of rainfall (fresh water) per hour, with wind at 20 miles per hour.

3.9.5 Shock; bench-handling. - The Converter, CV-146( )/TMQ-5, Power Supply, PP-968( )/TMQ-5 and Recorder Frequency-Time, RD-88( )/TMQ-5 shall meet full specified performance and there shall be no physical damage.

3.10 Tropicalization. - In addition to the requirement of MIL-P-11268 and the equipment drawings the following is required:

3.10.1 Statement of treatment. - The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use. (See 6.1i).

3.10.2 Salt fog. - After testing as specified in 4.14, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.10.3 Fungus. - The equipment (including accessories) shall show no evidence of visable fungus or corrosion when subjected to the test specified in 4.15. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

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3.11 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.5)

3.12 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery.) (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

d. (1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.16)

4.3 Preproduction inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C. (See Tables I and II, respectively)

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4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

ITEM	Where Required
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to bounce preconditioning (See 3.11) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce pre-conditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government - - at its discretion - - may perform all or any part of the specified requirements. (See 6.4) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table I. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table I. Group A inspection shall be performed in any order which is satisfactory to the Government except the operational inspection shall be last.

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TABLE I - GROUP A INSPECTION

Inspection	Req Par	Test Par	AQL Major	AQL Minor
Visual and mechanical				
Cabinet CY-1390( )/TMQ-5	3.12	4.8)		
Cabinet MX-1482( )/TMQ-5	3.12	4.8)		
Case CY-930( )/TMQ-5	3.12	4.8)	1%	4%
Converter CV-146( )/TMQ-5	3.12	4.8)		
Panel, Control C-834( )/TMQ-5	3.12	4.8)	COMBINED FOR THE GROUP	
Power Supply PP-968( )/TMQ-5	3.12	4.8)		
Recorder RD-88( )/TMQ-5	3.12	4.8)		
Paper, Graph ML-432( )/TMQ-5	3.8	4.8)		
Electrical *and mechanical operational				
Recorder, Frequency-line RD-88( )/TMQ-5				
Signal Voltage range	3.7.1.1	4.9.1.12)		
Speed of response	3.7.1.2	4.9.1.1)		
Linearity	3.7.1.3	4.9.1.2)		
Stability	3.7.1.4	4.9.1.3)		
Hysteresis	3.7.1.5	4.9.1.4)	1%	4%
Zero Adjustment	3.7.1.6	4.9.1.5)	COMBINED FOR THE GROUP	
Chatter	3.7.1.7	4.9.1.6)		
Pen lift	3.7.1.8	4.9.1.4)		
Pen Pressure	3.7.1.9	4.9.1.7)		
Chart Feed	3.7.1.10	4.9.1.8)		
Torque	3.7.1.11	4.9.1.9)		
Converter, Signal Data CV-146( )/TMQ-5				
Audio Frequency Range	3.7.2.1	4.9.1.12)		
Reference adjust range	3.7.2.2	4.9.1.10)	1%	4%
Linearity	3.7.2.3	4.9.1.2)	COMBINED FOR THE GROUP	
Stability	3.7.2.4	4.9.1.3)		

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TABLE I GROUP A INSPECTION- CONT'D

Combined requirements of RD-88( )/TMQ-5 and CV-146( )/TMQ-5			
Hysteresis	3.7.3.1	4.9.1.4)	
Linearity	3.7.3.2	4.9.1.2)	1% 4%
Stability	3.7.3.3	4.9.1.11)	
Audio Signal Voltage & waveform	3.7.3.4	4.9.1.12)	COMBINED FOR THE GROUP
Power Supply PP-968( )/TMQ-5			
Outputs	3.7.4.1	4.9.2	1%
Miscellaneous			
Speaker	3.7.6	4.9.4)	1% 4%
Fan operation	3.7.5	4.9.3)	COMBINED FOR THE GROUP

*All electrical defects are considered major.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to pars. 3.6 and 4.7 and to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection. Group B inspection shall normally be performed on inspection lots that have been subjected to and met the group A inspection.

4.5.3 Group C inspection. - This inspection shall consist of the tests specified in Table II and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1.

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TABLE II - GROUP C INSPECTION

Inspection	Req Par	Test Par
Sub Group I		
Temperature	3.9.2	4.10
Moisture-resistance test for equipment	3.9.3	4.11
Rain	3.9.4	4.12
Sub Group II		
Shock; bench-handling	3.9.5	4.13
Salt fog	3.10.2	4.14
Fungus	3.10.3	4.15

4.5.3.1 Sampling for Inspection of equipment. - Production sampling for Group C inspection shall be three (3) items selected from the first ten (10) items produced. Two (2) of these items shall be subjected to Sub Group I inspection and the third item shall be subjected to Sub Group II inspection. One (1) item shall be selected from the first three (3) items produced in the second half of production and this item shall be subjected to Sub Group I inspection. The item shall be selected at random. When preproduction samples are specified by contract or purchase order Group C inspection of samples from the first ten (10) items will not be required.

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the process and materials, as applicable, and, all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample units failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

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4.5.4 Reinspection of Conforming Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Bounce preconditioning. - The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for five (5) minutes. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.8 Visual and mechanical inspection of equipment. - The equipment shall be examined for the defects listed in MIL-STD-454 and Table III.

TABLE III - Classification of visual and mechanical defects

Classification	Defect
Major	Improper loose fitting panels and covers. Controls loose or too tight. Pen lifting and/or sliding action not smooth due to mechanical defects.

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TABLE III - Classification of visual and mechanical defects - Cont'd

Classification	Defect
Major	Improper meshing of gears (Misalinement.) Paper does not unroll smoothly and evenly Cabinet jacks and chassis plugs not properly alined. Vent in top of case will not open Fan not centered in opening Paper improperly or poorly printed Holes in paper improperly spaced or alined Components not interchangeable Marking assembly difficult, or impossible to reposition on drive cable.
Minor	Loose control knobs Shock mounts improperly mounted, or loose Speaker not centered in opening Panel handles improperly alined Frequency-Time Recorder RD-88( )/TMQ-5 view-window misalined or loose Loose chart roller and/or guide rods and bearings Vent in top of case loose and/or will not stay closed Loose and/or tight fitting parts in case CY-930( )/TMQ-5

4.9 Performance tests. - Test equipment shall conform to MIL-C-45662. A Recorder, Radiosonde AN/TMQ-5( ) shall be GLE to the successful bidder for test pruposes when the Converter, Signal Data CV-146( )/TMQ-5, Power Supply PP-968( )/TMQ-5 or the Recorder, Frequency-Time RD-88( )/TMQ-5 is procured as a separate item.

4.9.1 Standard test conditions. - Except where otherwise specified, the test shall be performed under the following standard test conditions:

a. Conduct test at normal room temperature and humidity after practicable thermal equilibrium has been attained.

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b. Allow the equipment to warm up for 30 minutes in steady stand-by position before taking measurements.

c. Adjust operating controls to permit optimum operation of the equipment; unless otherwise specified, operate on 115 volt 60-Hz current.

d. Connect the signal input cable to Frequency Standard TS-65C/FMQ-1, or equal.

e. Set the selector switch to "Signal".

f. Apply a 190 Hz . signal to the signal input.

4.9.1.1 Speed of response. - With a line voltage of 105 volts and under the conditions of Table IV the pen-sweep shall determine compliance with 3.7.1.2 in the times indicated.

TABLE IV

Line Frequency	Temperature	Signal Level	Speed of Response
50	Room	100 V Sawtooth	2-3/4 Seconds Maximum
50	Room	10-30 V Sawtooth	2-1/2 Seconds Maximum
50	-40°	10-30 V Sawtooth	2-3/4 Seconds Maximum
50	-40°	100 V Sawtooth	3 Seconds Maximum
60	Room	100 V Sawtooth	2-1/2 Seconds Maximum
60	-40°	10-30 V Sawtooth	2-1/2 Seconds Maximum
60	-40°	100 V Sawtooth	2-3/4 Seconds Maximum

4.9.1.2 Linearity test. - Using the frequency standard (item 4.9.1(d)), determine compliance of the recorder with 3.7.1.3 and 3.7.3.2.

4.9.1.3 Stability test. - Determine whether recorder conforms to 3.7.1.4 and 3.7.2.4 at input frequencies of 190 Hz, 100 Hz and 10 Hz.

4.9.1.4 Hysteresis and pen-lift test. - The input signal shall be varied from 10 Hz through 220 Hz and back to 10 Hz. The record shall conform to 3.7.1.5, 3.7.1.8 and 3.7.3.1.

4.9.1.5 Zero Adjust Test. - With "SIGNAL SELECTOR" switch set at "S.C." adjust "REC. ZERO" control; the range of pen movement shall determine compliance of 3.7.1.6.

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4.9.1.6 Chatter test. - With selector switch in S.C. position depress the "REC TEST" switch and hold until the pen comes to rest, then release "REC TEST" switch and allow pen to return to zero position to determine compliance with 3.7.1.7.

4.9.1.7 Pen pressure. - The pen pressure shall conform to 3.7.1.9.

4.9.1.8 Chart feed check. - During the course of the following tests, note the uniformity and accuracy of the chart feed when the equipment is operated from a 60 Hz power source. The tests shall conform to 3.7.1.10.

4.9.1.9 Torque Tests. - The main drive gear torque and the recorder mechanism drag shall be measured for compliance with 3.7.1.11.

4.9.1.10 Recording corresponding to 190 Hz. - Observe the position of the recording on the chart. The recording shall be at 95 chart divisions and shall be adjustable to this position in accordance with 3.7.2.2.

4.9.1.11 Constant output test with fluctuating power source. - Use all nine combinations of line voltage 105, 115, and 125 and the line frequencys 50, 60, and 65 to determine compliance with 3.7.3.3. This test may be made in conjunction with 4.9.1.3.

4.9.1.12 Signal voltage waveform. - The following tests shall be performed every 20 Hz starting at 10 Hz.

4.9.1.12.1 Comparison of response of negative going sawtooth and pulse signals. - Using negative going sawtooth and pulse signals of known frequency, vary the amplitude of the input signals and determine compliance with 3.7.3.4.

4.9.1.12.2 Comparison of responses to sine-wave, sawtooth and pulse signals. - Using sine-wave, sawtooth and pulse signals, all of the same frequency, determine compliance with 3.7.3.4.

4.9.2 Power Supply PP-968( )/TMQ-5. - The power supply shall be tested to determine compliance with 3.7.4.

4.9.3 Fan Operation Test. - While undergoing the performance tests listed above check the fan operation. Place a piece of paper over the fan opening; it shall remain in place when the fan is in operation.

4.9.4 Speaker output test. - The quality of the signal from the speaker shall conform to 3.7.6.

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4.10 Temperature test for equipment. - The equipment shall be subjected to the temperature cycle shown in MIL-STD-169. Measurements shall be taken as follows:

Step	Para.	Performance
1)	4.9)	Full Specification Requirements
4)	3.9.2)	
8)	3.9.2)	
10)	4.9)	

NOTE: Following exposure at -80°F and preceding step 8, the equipment shall be operated on stand-by power or conditioned at normal room temperature and humidity for 6 hours.

4.10.1 Failure. - If the equipment fails to meet the performance established by Step 1 during the other steps where measurements are taken, it shall be realined or readjusted once. If the equipment then fails to meet the performance established in Step 1 or fails subsequently during cycling, it does not pass the test.

4.11 Moisture-resistance test for equipment. - The equipment shall be tested as follows:

4.11.1 Test conditions. -

a. Do not move components from the humidity chamber for measurements.
b. Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.11.2 Test method. -

a. Dry at 130° ±5°F for 24 hours.
b. Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.
c. Take measurements as specified in 4.11.3, and readjust or realine as necessary to meet full specification requirements.
d. Subject to continuous cycling for two 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. The measurements specified in 4.11.3 shall be made during the periods shown in the Standard and shall comply with 4.11.3.
e. After cycling has been completed, condition the equipment for 6 hours, at 77° ±5°F and 40 to 60 percent relative humidity; then adjust for optimum performance, using only those means provided by the components. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specification requirements for those measurements specified in 4.11.3.

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4.11.3 Performance. - Measurements made during test, and specified performance during cycling, shall be as follows:

Measurement	Reference Paragraph	Performance During Cycling
Linearity	4.9.1.2)	Full Specification Requirements
Hysteresis	4.9.1.4)	
Pen Lift Operation	4.9.1.4)	
Chart Feed	4.9.1.8)	
Chatter Test	4.9.1.6)	
Fan Operation	4.9.3)	

4.11.4 Failure. - If any component fails to meet the performance specified in 4.11.3 during cycling, it shall be realined or readjusted once. If the components then fails to meet 4.11.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements as specified in 4.11.2(e), it does not pass the test.

4.12 Rain test. - The component shall be tested, step by step, as follows:

4.12.1 Dry at 150° ±5°F for 48 hours.

4.12.2 Condition for 4 hours at 77° ±5°F and 40 to 60 percent relative humidity.

4.12.3 Take pretest data. The equipment shall meet full specification performance.

4.12.4 Place the equipment, in its normal operating position and the vent closed, in a test chamber maintained at 77° ±5°F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ±5°F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for one hour.

4.12.5 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ±5 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the component. Maintain this condition for one hour.

4.12.6 Rotate the component 90 degrees horizontally, to expose a second side of the component to the wind and rain. Maintain this condition for one hour.

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4.12.7 Continue as in 4.12.6, until all four sides of the equipment have been exposed to the wind and rain, each side for one hour.

4.12.8 Immediately after completion of the foregoing, repeat the measurement specified in 4.11.3 without removal from the chamber but with the rain and wind discontinued. Complete the measurements within one hour. The measurements shall meet specified performance.

4.12.9 Open the component. There shall be no visible evidence of leakage.

4.13. Shock test; bench-handling. - The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance.

4.14 Salt fog test. - The equipment (including accessories), in its normal operating position shall be tested in accordance with Method 509, Procedure 1, of MIL-STD-810. The equipment shall not be opened. Operation is not required. No rust shall be imposed on the test item by the test facility.

4.15 Fungus test. - The equipment (including accessories), shall be tested in accordance with Method 508, Procedure 1, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required.

4.16 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified. (See 6.1(c)).

5.1.1 Level A. -

5.1.1.1 Cleaning. - All components of Radiosonde Recorder AN/TMQ-5 shall be cleaned in accordance with the applicable process of MIL-P-116.

5.1.1.2 Drying. - Recorder, Radiosonde AN/TMQ-5( ) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

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5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Spares, accessories and ancillary items. - Spare parts, accessories and ancillary items, as required, shall be packaged as follows:

5.1.1.4.2.1 Electron tubes. - Spare tubes, and all tubes not securely clamped in place, shall be packaged individually in accordance with MIL-E-75.

5.1.1.4.2.2 Cable and wire rope assemblies. - Each assembly shall be packaged method 111. Wind each item into a coil of proportionate dimensions. Tie in three places with twine, conforming to T-T-871, type 1, 6 ply; secure the ends with ties.

5.1.1.4.2.3 Contact Electrical, Pen Recorder, Computer Humidity-Temperature CP-223( )/UM, and Paper Graph ML-432( )/TMQ-5( ). Each item shall be packaged method 1C-1.

5.1.1.4.2.4 Hand tools. - Each item shall be packaged in accordance with the applicable requirements of MIL-H-15424.

5.1.1.4.2.5 Other items. - Each item shall be packaged method 111. Cushion each item by wrapping in paperboard conforming to PPP-P-291, type 111. Secure the wrap with tape conforming to PPP-T-45, type 111.

5.1.1.4.3 Consolidated package. - Consolidate the items, packaged as specified in 5.1.1.4.2, within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125, designed to fit within the accessory case. Seal the top and bottom flaps with tape, not less than two inches wide, conforming to PPP-T-76. Stow the sealed box within the case.

5.1.1.4.4 Case, Accessories CY-930( )/TMQ-5. - Each case, together with the stowed consolidated package specified in 5.1.1.4.3, shall be packaged method 111. Secure the cover with the fasteners provided. Cushion each case on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification. In addition, seal all seams and joints with tape, not less than two inches wide, conforming to PPP-T-76.

5.1.1.4.5 Cabinet, Electrical Equipment CY-1390( )/TMQ-5. - Each cabinet, consisting of the respective components installed and secured, shall be packaged method 111 in accordance with the procedure specified in 5.1.1.4.4.

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5.1.2. Level C. - Each Recorder, Radiosonde AN/TMQ-5( ) shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be level A, B or C as specified. (See 6.1c)

5.2.1 Level A. - Each Recorder, Radiosonde AN/TMQ-5( ), unit packaged as specified in 5.1, shall be packed within a close-fitting wood box conforming to PPP-B-601, overseas type, or PPP-B-621, class 2, style 7. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Closure and strapping conforming to QQ-S-781, type 1, Class B, shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B. - Each Recorder, Radiosonde AN/TMQ-5( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C. - Each Recorder, Radiosonde AN/TMQ-5( ), packaged as specified in 5.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type required.

c. Level of preservations and packing. (See Section 5)

d. When the rough handling and functional tests are required. (See Section 5)

e. The specific paragraphs of Section 5 which are applicable to the particular procurement.

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f. Preproduction inspection:

(1) Three sample units of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on two sample units while complete performance measurements can be made on the third sample unit. (See 3.2)

g. Marking and shipping of samples.

h. Place of final inspection.

i. Submission of the statement of treatment referenced in 3.10.1, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

j. See paragraph 4.9 for GLE requirements.

6.2 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example AN/TMQ-5W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1,)

6.3 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.4 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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PREPARING ACTIVITY
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PROJECT NUMBER
6660-A016

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-10882D(EL), RECORDER, RADIOSONDE AN/TMQ-5( ) (Units of)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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